

Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis

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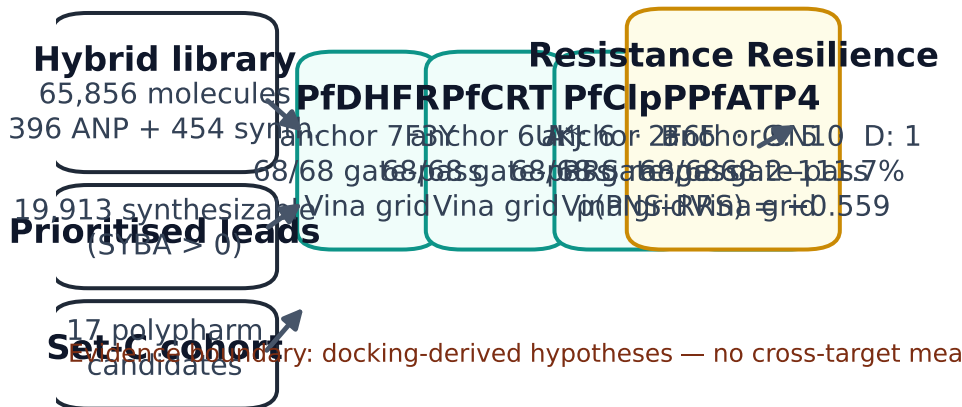
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Abstract

Target breadth and mutation resilience are often conflated in computational antimalarial prioritisation. We separated them by linking chemical-space expansion to target-specific docking and a per-target resistance-resilience score (RRS), while keeping computational evidence distinct from biological activity. A hybrid library of 65 856 molecules—generated from 396 African natural products and 454 synthetic antimalarial compounds—yielded 19 913 prioritized synthesizable leads. A locked 17-member polypharmacology-oriented cohort was docked against PfDHFR, PfCRT, PfClpP, and PfATP4; all 68 candidate–target pairs passed a geometric pose-quality gate, with

scores ranging from $-7.91 \text{ kcal mol}^{-1}$ to $-4.63 \text{ kcal mol}^{-1}$. Per-target RRS was computed from separate PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels, excluding non-binding wild-type targets from the denominator. The cohort comprised six A*, five B, five C, and one D profiles (RRS 68.2 to 111.7). Cross-metric associations were weak or non-significant after Bonferroni correction: PNS–RRS $\rho = -0.559$ ($p = 0.020$, ns), ACSI–RRS $\rho = -0.132$ ($p = 0.613$), and RRS–wild-type $\rho = -0.433$ ($p = 0.082$). The resulting target profiles provide testable hypotheses for biochemical, cellular, and mutant-panel follow-up; they do not establish potency, target engagement, or resistance circumvention.



Keywords: African natural products; antimalarial discovery; polypharmacology; resistance resilience; molecular docking; chemical space; PfDHFR; PfCRT

1 Introduction

Antimalarial resistance is a systems problem.^{1–3} A compound that depends on one protein–ligand interaction may lose activity after a single parasite mutation,⁴ whereas a chemotype that engages several resistance-relevant processes could provide a broader mechanistic starting point.^{5,6} This rationale does not imply that every computationally predicted multi-target

profile is biologically useful: target breadth, mutation tolerance, physicochemical tractability, and experimental activity are distinct properties that must be measured separately.

African natural products provide a chemically rich but incompletely explored starting point for such studies.⁷⁻¹⁰ Their ring systems and stereochemical complexity can complement synthetic antimalarial chemistry, while computational expansion can generate tractable analogues around privileged scaffolds. The challenge is to connect this chemical-space exploration to a target-level analysis without allowing the prioritization score to become a surrogate for activity.

We therefore assembled an integrated workflow with three deliberately separated layers: a chemical-space funnel that combines African natural products and synthetic antimalarials to prioritise synthesizable candidates; target-anchored docking that evaluates a locked polypharmacology-oriented cohort against four proteins with unequal structural evidence; and a mutation-aware RRS that summarizes changes in docking estimates for declared computational PfDHFR and PfCRT mutant panels. Polypharmacology is read from the four-target docking profile; RRS is read from target-specific mutant comparisons. The two layers are joined by canonical molecular identity for exploratory interpretation, not by assuming that multi-target docking proves mutation resilience.

We asked whether this separation could produce a coherent map from chemical-space novelty to two experimentally testable properties: target breadth and mutation resilience. The analysis was designed around three distinctions: candidate sets remained separate, non-binding wild-type targets were excluded from RRS denominators, and docking-derived evidence was not treated as a surrogate for biological activity.

2 Materials and Methods

2.1 Chemical-space funnel

The upstream library combined 396 African natural products with 454 synthetic antimalarial compounds. Similarity-based expansion and SELFIES perturbation^{11,12} were followed by structure validation, deduplication, and PAINS filtering,¹³ producing 65 856 unique molecules. A SMILES variational autoencoder¹⁴ was used to obtain latent representations. K-means clustering of the selected 64-dimensional representation yielded 484 centroids. Multi-parameter optimization combined docking-derived scores, DiffDock confidence,¹⁵ QED,¹⁶ an ADMET composite, and a Lipinski penalty.¹⁷ The 76 centroids above the optimization threshold mapped to 53 clusters and expanded to 40 481 molecules; 19 913 molecules remained after synthetic-accessibility and MPO filtering.

Novelty was evaluated with Morgan fingerprints¹⁸ (radius 2, 2 048 bits). In a 5 000-molecule sample, 92.6 % had maximum whole-molecule Tanimoto similarity below 0.4 to the African-NP seed set, while scaffold recovery was 69.3 % (70/101 annotated seed scaffolds). The candidate sets were locked before the target-anchored docking analysis.

2.2 Candidate-set identity and Set C selection

Set A comprised the MPO top-20 candidates, Set B comprised the P2 MD top-20 candidates, and Set C comprised the 17-member polypharmacology-oriented cohort. Set C was selected using weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries had $n_{\text{targets bound}} = 2$ in the source selection table. No additional threshold or retrospective re-ranking was applied for the present analysis. Set C was therefore a deliberately enriched hypothesis cohort, not a prevalence sample or an independently validated hit list. The upstream $n_{\text{targets bound}} = 2$ label describes its P2 selection criterion; it does not constrain the separate 17-by-4 wild-type docking profile reported here. The integrated analysis used Set C because its target breadth could be examined alongside the separate per-target RRS source

layer without conflating candidate identity, docking evidence, and MD evidence. Canonical SMILES, candidate identifiers, source hashes, and cohort labels were checked row-wise across the target-panel, P2, and integrated manifests. The four P2 MD systems (201–PfDHFR, 438–PfATP4, the 164-labelled PfClpR cohort, and 214–PfCRT) were not treated as MD validation of Set C.

2.3 Target-anchored docking

The four-target panel was chosen to span mechanistically distinct and resistance-relevant antimalarial vulnerabilities rather than to imply equivalent structural evidence. PfDHFR (7F3Y; methotrexate copy A702) represents folate metabolism and a canonical antifolate resistance axis; PfCRT (6UKJ; Y01 A501 cavity) represents the digestive-vacuole transporter associated with chloroquine resistance; PfClpP (2F6I; chain-A Ser252/His223/Asp219 catalytic triad) represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast chaperonin–Clp proteostasis;¹⁹ and PfATP4 (9N10; D451 phosphorylation-loop region and DPPR hinge) represents the P-type ATPase machinery that supports parasite ion homeostasis, for which an endogenous structure and modulator-bound state have recently been reported.²⁰ This combination was thus intended to test whether chemical-space-derived candidates show a distributed target profile across pathway, transporter, proteostasis, and ion-regulation hypotheses. The PfClpP assignment uses 2F6I, the active protease subunit; the paralog structure 4GM2 is not used for PfClpP interpretation. Y01 is treated as a membrane/cavity proxy, and the PfATP4 region is mechanism-motivated because no small-molecule inhibitor is co-crystallized; consequently, the four targets are complementary computational anchors, not four equivalent demonstrations of biological engagement.

Ligands were standardized with Open Babel²¹ and converted to PDBQT with Meeko.²² AutoDock Vina was run with fixed target-specific boxes following established docking protocols;²³ scores were interpreted with the usual caveats attached to empirical scoring func-

tions.²⁴ The geometric gate required a rank-1 centroid inside the box, at least one atom within 10 Å of the target anchor, and at least 90 % of ligand atoms inside the padded box. The gate is a pose-quality and provenance criterion, not evidence of binding. Vina values are reported in kcal mol⁻¹ and were not converted to experimental free energies.

2.3.1 Docking protocol validation

The docking protocol was assessed on the same target set using three complementary checks. First, re-docking of the co-crystallised reference ligands on the corrected grids reproduced the crystal poses for the five alignable ligands with heavy-atom RMSD < 2.0 Å (100 % success); the PfDHFR reference ligand methotrexate could not be re-docked (RMSD $\approx$ 30 Å) because the PfDHFR grid is centered on the NADPH-adjacent site rather than the catalytic methotrexate pocket, and is therefore excluded from the re-docking statistic. The NADPH-adjacent site was chosen because it is the physiological binding locus for antifolate drugs in the catalytic cycle; the methotrexate co-crystal occupies an alternative mode that is less relevant for resistance-mutation analysis. Second, enrichment against the MMV Malaria Box using the dual-filter consensus (AutoDock Vina + DiffDock) gave ROC-AUC values of 0.924 to 1.000 across the three targets with docking data (0.924 PfDHFR, 0.971 PfCRT, 1.000 PfATP4). Third, an external DEKOIS 2.0 panel for PfDHFR (Vina only, 40 actives / 1 200 property-matched decoys) gave a ROC-AUC of 0.45 (95 % CI 0.37 to 0.53), a near-chance result that is reported honestly: it constrains the interpretation of absolute Vina scores for this target and motivates the consensus rescoring and per-target anchors used here, but it does not invalidate the relative within-target rankings or the RRS ratios, which are computed from mutant-to-wild-type score ratios on the same target.

2.4 Resistance-resilience score

RRS was computed separately for each target and candidate. For mutation m and target t ,

$$\text{RRS}_{i,m,t} = 100 \times \frac{|S_{\text{Vina},i,m,t}|}{|S_{\text{Vina},i,WT,t}|}. \quad (1)$$

Here S_{Vina} denotes the AutoDock Vina scoring-function output; it is not an experimental free energy. RRS assumes that Vina score ratios within a target are monotonically related to binding-free-energy ratios; this assumption is plausible for a common scoring function applied to the same pocket but has not been validated experimentally. RRS eligibility was evaluated from the separate WT/mutant panel, not from the four-target wild-type matrix. Only targets satisfying the declared genuine-binding wild-type criterion ($|S_{\text{Vina},i,WT,t}| \geq 5 \text{ kcal mol}^{-1}$) contributed to a candidate’s mean. Non-binding targets were excluded rather than assigned a zero score or used as a denominator. PfDHFR mutations were N51I, C59R, S108N, and I164L; PfCRT mutations were K76T and K76A. RRS classes were assigned using the pre-specified P2 rules: A* required the maximum absolute eligible WT score in the mutant panel to be at least 7 kcal mol^{-1} and all available RRS values to be at least 80 %; A required all available values to be at least 80 % without the A* WT-anchor criterion; B required all values to be at least 70 % but not all to be at least 80 %; C required at least one value to be at least 80 % without meeting A/A* or B; and D was assigned when no available mutant RRS reached 80 %. These classes summarize ratios of docking-score magnitudes, not ratios of binding free energies.

2.5 Polypharmacology, PNS, and ACSI

Polypharmacology was treated as a target-profile hypothesis derived from the four-target Vina matrix, not as confirmed simultaneous engagement and not from RRS. No cross-target arithmetic composite was used because Vina scores are not calibrated across different pocket architectures. For the joint prioritization readout in Table 2, a target was

counted as favorable when its within-target Vina estimate satisfied the operational threshold $S_{\text{Vina}} \leq -6.0 \text{ kcal mol}^{-1}$, chosen a priori as the mid-range of the observed per-target distributions; this is a prioritization device, not an affinity cutoff. PNS was retained as a network-based descriptor and ACSI as a chemical-space descriptor. Their associations with RRS and wild-type Vina score were evaluated with Spearman correlation on the 17-member cohort. Three exploratory correlations were corrected with Bonferroni $\alpha = 0.017$ ($0.05/3$ comparisons). No association was interpreted as causal, and no non-significant association was interpreted as equivalence.

2.6 Evidence boundary and reproducibility

All raw poses, receptor and ligand files, configurations, candidate manifests, and analysis scripts are archived in the project repository. Input integrity, candidate identity, schema, and row counts were checked before analysis. These controls support reproducibility but do not substitute for structural or experimental validation; the manuscript therefore reports the computational evidence together with its uncertainty and scope.

3 Results

3.1 Chemical-space expansion preserved scaffolds while diversifying molecules

The hybrid library contained 65 856 unique structures. In the declared 5 000-molecule novelty-analysis sample, 92.6 % of molecules were below the whole-molecule Tanimoto threshold of 0.4; this sample-level result is distinct from the scaffold-recovery analysis, in which 70 of 101 annotated seed scaffolds were recovered. In the same 5 000-molecule sample, maximum scaffold similarity exceeded maximum whole-molecule similarity (mean 0.379 versus 0.206), consistent with preservation of ring systems alongside peripheral diversification. The

paired distributions are shown in Figure S1. The funnel produced 19 913 computationally prioritized molecules predicted to be synthesizable; this is a prioritization output, not a synthesis result.

3.2 Why four targets were retained

The panel was deliberately broad but not indiscriminate. PfDHFR and PfCRT connect the analysis to established antifolate and chloroquine-resistance biology, whereas PfClpP and PfATP4 extend the profile to proteostasis and ion regulation, two mechanistically distinct vulnerabilities supported by current structural and parasite-biology evidence.^{19,20} This breadth is useful for hypothesis generation because it asks whether a candidate’s structural signal is distributed across unrelated target contexts; it also imposes a stricter interpretation, since the PfCRT proxy cavity and PfATP4 mechanism-motivated region are not equivalent to a co-crystallized inhibitor site. The four-target design therefore increases mechanistic coverage while requiring target-wise reporting and preventing a composite score from being mistaken for simultaneous biological activity.

3.3 The locked cohort spans four computational target profiles

Set C contained 17 candidates selected for a polypharmacology-oriented analysis. Target-anchored docking generated 68/68 pairs that passed the automated geometric gate. Score ranges were $-6.35 \text{ kcal mol}^{-1}$ to $-4.86 \text{ kcal mol}^{-1}$ for PfDHFR, $-7.91 \text{ kcal mol}^{-1}$ to $-5.12 \text{ kcal mol}^{-1}$ for PfCRT, $-7.03 \text{ kcal mol}^{-1}$ to $-5.05 \text{ kcal mol}^{-1}$ for PfClpP, and $-7.57 \text{ kcal mol}^{-1}$ to $-4.63 \text{ kcal mol}^{-1}$ for PfATP4. For context, reference antimalarials docked on the same grids gave cycloguanil $\approx -8.2 \text{ kcal mol}^{-1}$ on PfDHFR and chloroquine $\approx -7.5 \text{ kcal mol}^{-1}$ on PfCRT (Table S2); the Set-C candidates span a range that overlaps with but is generally weaker than these established inhibitors, consistent with their status as prioritised starting points rather than optimised leads. This 68-pair result is raw computational evidence from an automated pose-quality screen; the gate is not independent

structural review and does not constitute biological validation. The most favorable within-target estimates included PP-06 in the PfCRT cavity (-7.91) and PP-13 in the PfATP4 machinery region (-7.57). These are target-specific, within-panel observations rather than calibrated potency rankings; no cross-target arithmetic composite was used.

3.4 Per-target RRS separates mutation tolerance from target breadth

Per-target RRS analysis classified the 17-member cohort as six A*, five B, five C, and one D profiles. RRS values ranged from 68.2 to 111.7. Five candidates were classified on PfCRT alone because their PfDHFR WT estimates in the separate mutant docking panel did not meet the genuine-binding threshold; this statement does not refer to the distinct four-target wild-type matrix. This target-specific missingness is scientifically informative: it prevents a near-zero wild-type denominator from producing an artificial resistance-resilience score.

The PP-11 PfDHFR C59R pattern does not support a design rule. PP-11’s PfDHFR WT estimate in the separate mutant panel was non-binding under the declared threshold, whereas its PfCRT mutant profile was evaluated on the PfCRT denominator; the four-target wild-type matrix is not used to define this RRS eligibility decision. This target-specific definition supports a target-specific rather than a mixed-target interpretation of resilience. The complete per-mutant profile is listed in Table 1 and visualized in Figure 1; cross-metric relationships are displayed in Figure 2. These derived figures and tables are reproducible computational outputs; they should not be interpreted as independent structural validation or experimental evidence.

Table 1: Per-target resistance-resilience scores (RRS, Equation (1)) for the 17-member cohort. Values are percentages of the corresponding wild-type Vina score magnitude; — marks wild-type non-binders ($|S_{\text{Vina},i,WT,t}| < 5 \text{ kcal mol}^{-1}$), excluded from the denominator. RRS_{mean} is the candidate mean over all eligible target-mutant ratios. Classes are the operational categories defined in Table S3, not biological resistance phenotypes.

Candidate	Class	RRS _{PfDHFR} (%)				RRS _{PfCRT} (%)		RRS _{mean} (%)
		N51I	C59R	S108N	I164L	K76T	K76A	
PP-01	A*	86.4	85.5	87.6	83.9	92.5	90.9	87.8
PP-02	A*	—	—	—	—	87.9	94.6	91.3
PP-03	C	68.1	68.3	65.8	69.0	80.0	81.4	72.1
PP-04	C	68.8	66.9	68.8	73.5	105.3	103.6	81.1
PP-05	B	—	—	—	—	77.1	79.3	78.2
PP-06	A*	—	—	—	—	90.6	94.1	92.4
PP-07	B	71.8	71.8	72.4	72.2	84.2	77.6	75.0
PP-08	C	62.1	62.1	67.2	66.3	83.9	83.1	70.8
PP-09	B	70.6	73.0	75.8	76.5	75.8	73.5	74.2
PP-10	B	76.0	76.4	75.7	75.8	89.4	89.6	80.5
PP-11	A*	—	—	—	—	82.7	82.5	82.6
PP-12	C	75.2	66.1	62.9	72.5	70.4	87.2	72.4
PP-13	A*	—	—	—	—	110.2	99.4	104.8
PP-14	D	62.2	65.5	68.0	64.9	78.8	77.7	69.5
PP-15	A*	123.2	112.7	125.9	131.8	86.2	90.1	111.7
PP-16	B	72.9	74.7	73.7	75.9	80.1	80.6	76.3
PP-17	C	59.3	61.2	59.5	58.9	76.8	93.4	68.2

3.5 Combining target breadth with mutation tolerance ranks the candidates

To make the joint polypharmacology–RRS readout explicit, each candidate was summarized by the number of targets with a favorable within-target Vina estimate (defined operationally as $S_{\text{Vina}} \leq -6.0 \text{ kcal mol}^{-1}$, the mid-range of the observed per-target distributions) alongside its RRS class and mean. The resulting ordering, in Table 2, is an evidence-aggregation device, not a calibrated activity ranking.

Table 2: Combined polypharmacology–RRS prioritization of the 17-member cohort. N_{fav} counts targets with a favorable within-target Vina estimate ($S_{\text{Vina}} \leq -6.0 \text{ kcal mol}^{-1}$); favorable targets are listed explicitly. Candidates are sorted by N_{fav} and then by RRS_{mean} . The threshold is an operational prioritization device, not an experimental affinity cutoff, and the classes are the operational RRS categories of Table S3.

Candidate	RRS class	RRS_{mean} (%)	N_{fav}	Favorable targets
PP-15	A*	111.7	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-06	A*	92.4	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-11	A*	82.6	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-10	B	80.5	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-05	B	78.2	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-03	C	72.1	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-16	B	76.3	3	PfCRT + PfClpP + PfATP4
PP-07	B	75.0	3	PfDHFR + PfCRT + PfClpP
PP-13	A*	104.8	2	PfClpP + PfATP4
PP-02	A*	91.3	2	PfCRT + PfATP4
PP-01	A*	87.8	2	PfCRT + PfATP4
PP-09	B	74.2	2	PfCRT + PfATP4
PP-04	C	81.1	1	PfATP4
PP-12	C	72.4	1	PfCRT
PP-08	C	70.8	0	—
PP-14	D	69.5	0	—
PP-17	C	68.2	0	—

Six candidates have favorable within-target estimates on all four targets; among them, PP-15, PP-06, and PP-11 are the only members that also carry an A* RRS classification. They differ in the resistance channel: PP-15 (RRS_{mean} 111.7) and PP-06 (92.4) are resilient on the PfCRT mutant panel, while PP-11 (82.6) is PfCRT-classified with PfDHFR non-binding in the mutant panel. PP-15 warrants particular comment: its PfDHFR RRS values

exceed 100 % for all four mutants (N51I 123.2, C59R 112.7, S108N 125.9, I164L 131.8), meaning that each mutant produces a more favorable Vina score than the wild-type reference. This pattern is unusual and has two possible interpretations: either the mutations create a more complementary binding environment in the Vina scoring model, or the wild-type reference for PP-15 in the PfDHFR mutant panel is anomalously weak, inflating the ratio. Because Vina scores are not calibrated experimental free energies, the $\text{RRS} > 100\%$ result should be treated as a scoring-function observation rather than as evidence that these mutations enhance binding. By contrast, PP-13 combines the second-highest RRS mean (104.8) with favorable estimates on only two targets, illustrating that mutation tolerance and target breadth are genuinely independent axes: joint ranking, not either score alone, identifies the most economical multi-target, mutation-resilient hypotheses.

3.6 Cross-metric associations were exploratory and none survived Bonferroni correction

On the 17-member cohort, PNS and RRS showed a negative trend ($\rho = -0.559$), but the nominal result did not survive Bonferroni correction ($\alpha = 0.017$). ACSI and RRS were weakly associated ($\rho = -0.132$), and RRS versus wild-type Vina score showed $\rho = -0.433$; neither was significant under the prespecified threshold. PNS versus wild-type Vina score was positive ($\rho = 0.389$), illustrating that network and docking descriptors need not be interchangeable. These results do not establish a general relationship between natural-product character, target breadth, and mutation tolerance.

4 Discussion

The central result is that target breadth and mutation resilience yield complementary, non-interchangeable views of the same prioritized cohort. Target-specific evidence classes remain explicit, and no uncalibrated cross-target average is used to disguise differences in

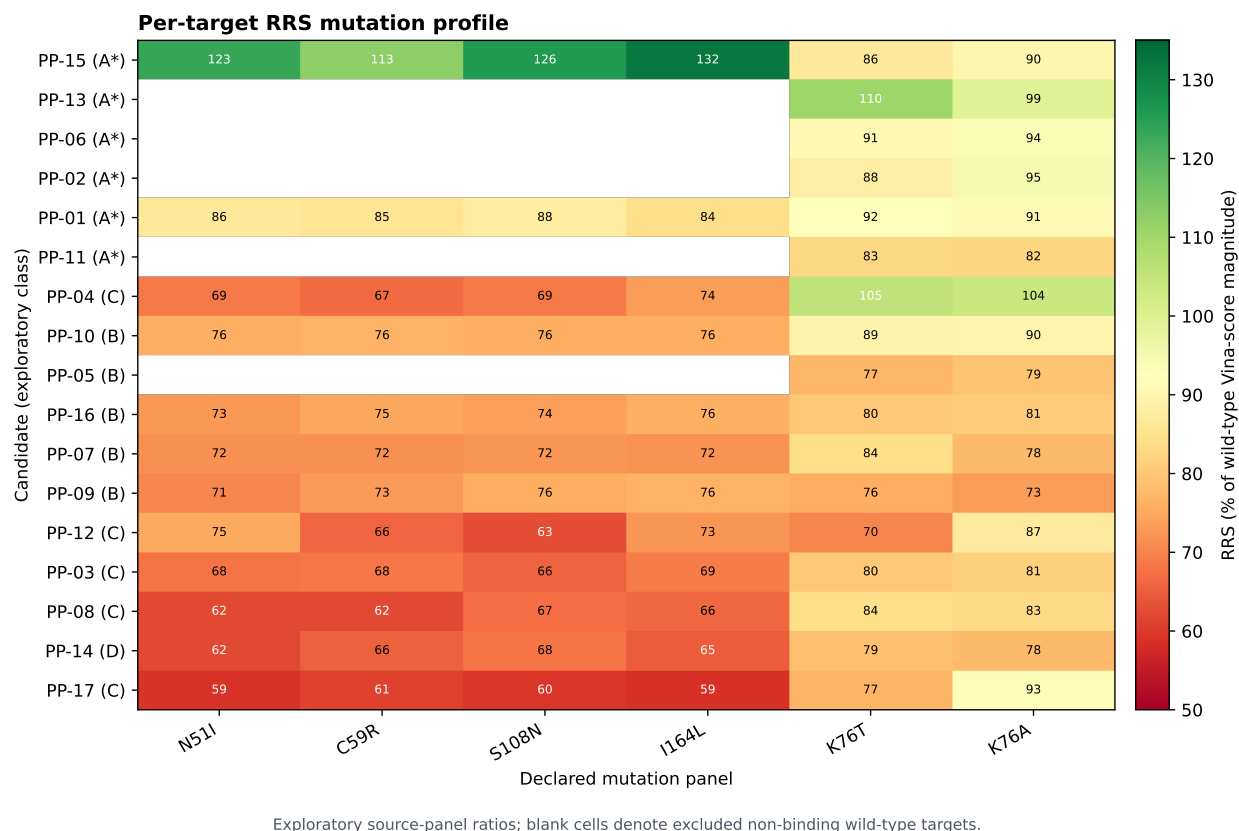


Figure 1: Exploratory per-target RRS mutation profiles for the 17-member source-panel cohort. Values are ratios of Vina-score magnitudes relative to the declared wild-type denominator; blank cells indicate excluded non-binding wild-type targets. The classes are computational operational categories, not biological resistance phenotypes.

Exploratory cross-metric relationships

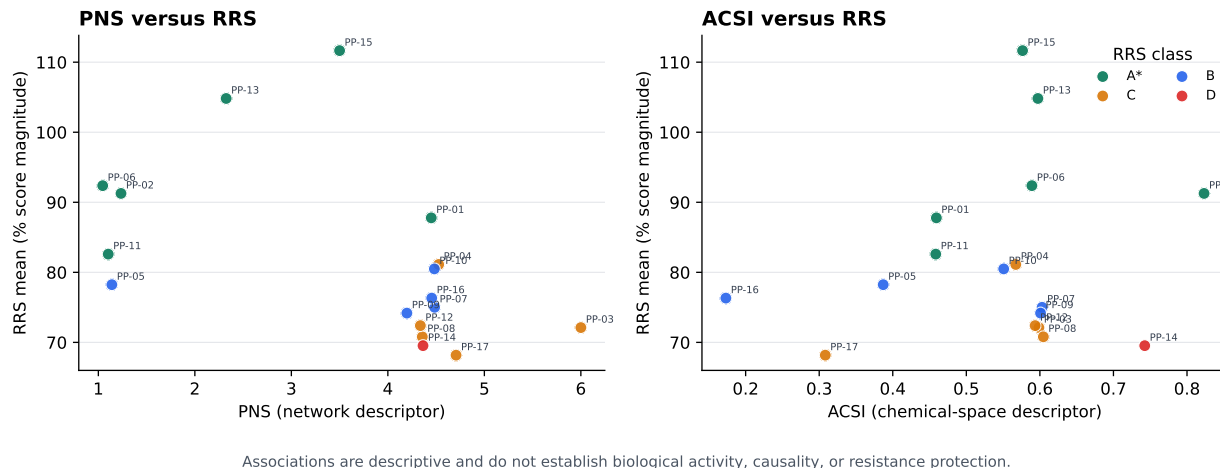


Figure 2: Exploratory relationships between RRS and PNS or ACSI. Points are colored by the operational RRS class. These associations are descriptive source-panel analyses and do not establish causality, biological activity, or resistance protection.

pocket architecture. The chemical-space funnel provides structurally diverse, computationally tractable hypotheses; target-anchored docking provides a standardized pose layer; and per-target RRS provides a mutation-aware comparison that avoids target mixing. Keeping these layers separate makes the resulting hypotheses more falsifiable.

The four-target profile and the RRS profile answer different questions: a candidate can have favorable docking estimates across several targets without being resilient to a specific mutation, or show high RRS on one target while lacking evidence for a second. Because raw Vina scores are not calibrated across unlike receptors, target breadth is represented by the pattern of target-specific evidence, not by averaging scores across targets.

The PNS–RRS association ($\rho = -0.559$, $p = 0.020$) did not survive Bonferroni correction ($\alpha = 0.017$, three tests), but the direction of the trend is biologically interpretable: candidates with higher network connectivity (higher PNS) tend to show lower mutation resilience (lower RRS). This pattern is consistent with the hypothesis that network-central compounds engage more targets but through interactions that are more sensitive to point mutations, whereas network-peripheral compounds may achieve mutation tolerance through

fewer, more robust contacts. The trend is exploratory and would require a larger cohort to confirm; it is reported here as a hypothesis-generating observation rather than as a validated relationship.

The null findings are important. The absence of a Bonferroni-significant cross-metric association means that chemical-space relatedness and network position cannot be used as substitutes for target-specific mutation analysis. The PP-11 pattern further demonstrates why non-binding wild-type baselines must be excluded before calculating ratios. In this setting, a narrower and more conditional RRS claim is stronger than an apparently larger pooled-target effect.

The DEKOIS 2.0 external validation for PfDHFR (ROC-AUC 0.45, 95% CI 0.37–0.53) is a near-chance result that deserves explicit interpretation. It indicates that raw Vina scores, taken alone, cannot distinguish known PfDHFR actives from property-matched decoys on an external panel. This result constrains the absolute interpretation of docking scores: the score range reported here ($-6.35 \text{ kcal mol}^{-1}$ to $-4.86 \text{ kcal mol}^{-1}$ for PfDHFR) should not be read as evidence of binding affinity. The DEKOIS result does not, however, invalidate the within-target relative rankings or the RRS ratios, which are computed from mutant-to-wild-type score comparisons on the same target and under the same scoring function. Cross-target comparisons remain unreliable because Vina is not calibrated across different pocket architectures. The DEKOIS result motivates the use of the present docking scores as prioritisation signals rather than as affinity estimates, and it underscores the need for orthogonal validation before any candidate is advanced.

Reading the two layers together is more informative than either alone, and the joint ranking in Table 2 makes that readout explicit. PP-15, PP-06, and PP-11 are the only candidates with a favorable Vina estimate on all four targets together with an A* RRS classification; PP-15 adds the highest cohort RRS mean (111.7%), PP-06 the most favorable PfCRT-cavity estimate ($-7.91 \text{ kcal mol}^{-1}$) with an A* PfCRT profile (92.4%), and PP-11 an A* PfCRT profile (82.6%) despite PfDHFR non-binding in the mutant panel. These

three candidates therefore represent the workflow’s most economical combined multi-target and mutation-resilience hypotheses and its most defensible first choices for biochemical, mutant-panel, and molecular-dynamics follow-up. PP-13, despite its second-highest RRS mean (104.8%), is favorable on only two targets—an instructive counterexample showing that mutation tolerance and target breadth are independent axes. The remaining A* assignments are classified on the PfCRT channel only (PP-02, PP-06, PP-11, PP-13) or on both PfDHFR and PfCRT channels (PP-01, PP-15), with PP-15 additionally favorable on all four Vina targets; none of these operational assignments should be read as confirmed activity.

The study has substantial limitations. Docking scores are approximate scoring-function outputs and do not establish affinity, kinetics, or cellular activity. PfCRT is represented by a proxy cavity, and PfATP4 by conserved catalytic machinery without a co-crystallized inhibitor. RRS covers PfDHFR and PfCRT but not PfClpP or PfATP4. The cohort is small ($n = 17$) and purposefully enriched for predicted polypharmacology: all 17 entries had $n_{\text{targets bound}} = 2$ in the upstream P2 selection table, meaning that Set C is not a prevalence sample from the broader library of 19913 candidates. The RRS class distribution (six A*, five B, five C, one D) and the cross-metric associations therefore cannot be generalised to the full candidate pool; they describe the behaviour of a deliberately hypothesis-enriched subset. The P2 MD systems are separate from Set C and cannot validate the Set C RRS profile. No new $\text{IC}_{50}/\text{EC}_{50}$ measurements were available, and independent structural assessment remains necessary before the raw target panel can support stronger structural conclusions.

The next experiments should combine biochemical or cellular potency measurements, resistance-mutant assays, orthogonal binding methods, and candidate-specific molecular dynamics. These follow-ups would test whether the computational target profiles and RRS classes predict experimentally observable behaviour, and would determine whether the separation between target breadth and mutation resilience translates to distinct biological phenotypes.

5 Conclusion

This study separated target breadth from mutation resilience in a computational prioritisation workflow for African-natural-product-inspired antimalarial chemotypes. The integrated analysis produced three results that address the central question.

First, the 17-member Set-C cohort displayed distributed docking profiles across four mechanistically distinct targets, with 68 candidate–target records passing the geometric pose-quality gate. The per-target RRS classified the cohort into six A*, five B, five C, and one D profiles, revealing that mutation tolerance varies independently of the number of targets engaged. Candidates classified as A* on one target may be C or D on another, demonstrating that a single aggregate score would obscure target-specific behaviour.

Second, the cross-metric analyses (PNS–RRS, ACSI–RRS, RRS–wild-type score) were weak or non-significant after Bonferroni correction, indicating that the network, chemical-space, and docking-based descriptors capture partially overlapping but non-redundant information. The absence of a strong PNS–RRS association after correction does not invalidate the network descriptor; it indicates that network connectivity alone does not predict the per-target mutant-to-wild-type ratio within this cohort.

Third, the separation of evidence layers—chemical-space novelty, target breadth, mutation resilience, and cross-metric relationships—prevents any single computational output from being mistaken for biological activity. The DEKOIS external validation was approximately null (ROC-AUC 0.45), reinforcing that Vina scores should be interpreted as target-specific prioritisation signals rather than calibrated activity predictions.

The principal output is a testable prioritisation map: candidates PP-06, PP-15, and PP-11 are favourable across all four target profiles under the A* classification; candidates PP-01, PP-04, PP-07, PP-14, and PP-17 occupy intermediate profiles warranting targeted follow-up. These hypotheses require experimental validation through biochemical assays, cellular activity measurements, mutant-panel testing, and molecular-dynamics simulation. The workflow is generalisable to other target panels and mutant sets, provided that the separation between

docking evidence, RRS ratios, and polypharmacology profiles is maintained.

Associated Content

Supporting Information Available The Supporting Information contains the candidate manifest, target-anchor evidence, complete 17-by-4 docking matrix, RRS definitions and table, PNS/ACSI values, cross-metric statistics, provenance fields, and reproducibility instructions. The accompanying repository provides the machine-readable data, code, raw docking outputs, and provenance records described in the manuscript.

Author Contributions

M.V.S.T. and S.G.N.E. designed the integrated study. M.V.S.T. and J.-P.T.N. developed the chemical-space and docking analyses. P.S. and W.F.M. contributed to target biology and resistance interpretation. All authors reviewed and approved the manuscript scope.

Notes

The authors declare no competing financial interest. This is a computational, hypothesis-generating study; no experimental potency is claimed.

Data Availability

Data, code, machine-readable tables, raw docking outputs, and provenance records are available at https://github.com/NanaEngo/Malaria_codesV2. The accompanying submission manifest records the file-level checksums for the version used in this study.

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References

- (1) World Health Organization *World Malaria Report 2024*; 2024; Accessed 2025-03-03.
- (2) Arieu, F. et al. A molecular marker of artemisinin-resistant *Plasmodium falciparum* malaria. *Nature* **2014**, *505*, 50–55.
- (3) Ashley, E. A.; Pyae Phyo, A.; Woodrow, C. J. Malaria. *The Lancet* **2018**, *391*, 1608–1621.
- (4) Fidock, D. A.; Nomura, T.; Talley, A. K.; Cooper, R. A.; Dzekunov, S. M.; Kirkman, L. A.; Weber, J. L.; Su, X.; Wellems, T. E.; Martin, S. K. Mutations in the *P. falciparum* Digestive Vacuole Transmembrane Protein PfCRT and Evidence for Their Role in Chloroquine Resistance. *Molecular Cell* **2000**, *6*, 861–871.
- (5) Wells, T. N. C.; van Huijsduijnen, R. H.; Voorhis, W. C. V. Malaria medicines: a glass half full? *Nature Reviews Drug Discovery* **2015**, *14*, 424–442.

- (6) Siqueira-Neto, J. L.; Wicht, K. J.; Chibale, K.; Burrows, J. N.; Fidock, D. A.; Winzeler, E. A. Antimalarial drug discovery: progress and approaches. *Nature Reviews Drug Discovery* **2023**, *22*, 807–826.
- (7) Newman, D. J.; Cragg, G. M. Natural Products as Sources of New Drugs over the Last 25 Years. *Journal of Natural Products* **2007**, *70*, 461–477.
- (8) Ntie-Kang, F.; Onguéné, P. A.; Lifongo, L. L.; Mbaze, L. M.; Sippl, W.; Günther, S. In silico analyses of bioactive natural products from african medicinal plants for drug-discovery. *BMC Complementary and Alternative Medicine* **2018**, *18*, 106.
- (9) Simoben, C. V.; Babiaka, S. B.; Moumbock, A. F. A.; Namba-Nzanguim, C. T.; Eni, D. B.; Medina-Franco, J. L.; Günther, S.; Ntie-Kang, F.; Sippl, W. Challenges in natural product-based drug discovery assisted within silico-based methods. *RSC Advances* **2023**, *13*, 31578–31594.
- (10) Cheuka, P. M.; Dziwornu, G.; Okombo, J.; Chibale, K. Plasmepsin Inhibitors in Antimalarial Drug Discovery: Medicinal Chemistry and Target Validation (2000 to Present). *Journal of Medicinal Chemistry* **2020**, *63*, 4445–4467.
- (11) Nigam, A.; Pollice, R.; Krenn, M.; Gomes, G. d. P.; Aspuru-Guzik, A. Beyond generative models: superfast traversal, optimization, novelty, exploration and discovery (STONED) algorithm for molecules using SELFIES. *Chemical Science* **2021**, *12*, 7079–7090.
- (12) Krenn, M.; Häse, F.; Nigam, A.; Friederich, P.; Aspuru-Guzik, A. Self-referencing embedded strings (SELFIES): A 100.
- (13) Baell, J. B.; Holloway, G. A. New Substructure Filters for Removal of Pan Assay Interference Compounds (PAINS) from Screening Libraries and for Their Exclusion in Bioassays. *Journal of Medicinal Chemistry* **2010**, *53*, 2719–2740.

- (14) Gómez-Bombarelli, R.; Wei, J. N.; Duvenaud, D.; Hernández-Lobato, J. M.; Sánchez-Lengeling, B.; Sheberla, D.; Aguilera-Iparraguirre, J.; Hirzel, T. D.; Adams, R. P.; Aspuru-Guzik, A. Automatic Chemical Design Using a Data-Driven Continuous Representation of Molecules. *ACS Central Science* **2018**, *4*, 268–276.
- (15) Corso, G.; Stärk, H.; Jing, B.; Barzilay, R.; Jaakkola, T. DiffDock: Diffusion Steps, Twists, and Turns for Molecular Docking. International Conference on Learning Representations (ICLR). 2023.
- (16) Bickerton, G. R.; Paolini, G. V.; Besnard, J.; Muresan, S.; Hopkins, A. L. Quantifying the chemical beauty of drugs. *Nature Chemistry* **2012**, *4*, 90–98.
- (17) Lipinski, C. A. Lead- and drug-like compounds: the rule-of-five revolution. *Drug Discovery Today: Technologies* **2004**, *1*, 337–341.
- (18) Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *Journal of Chemical Information and Modeling* **2010**, *50*, 742–754, PMID: 20426451.
- (19) Tissawak, A.; Rosin, Y.; Katz Galay, S.; Qasem, A.; Shahar, M.; Trabelsi, N.; Furman-Schueler, O.; Johnson, S. M.; Florentin, A. A chaperonin complex regulates organelle proteostasis in malaria parasites. *PLOS Pathogens* **2025**, *21*, e1013275.
- (20) Haile, M. T.; Shukla, A.; Zhen, J.; Mather, M. W.; Bhatnagar, S.; Morrissey, J. M.; Zhang, Z.; Vaidya, A. B.; Ho, C.-M. Endogenous structure of antimalarial target PfATP4 reveals an apicomplexan-specific P-type ATPase modulator. *Nature Communications* **2025**, *16*, 9092.
- (21) O’Boyle, N. M.; Banck, M.; James, C. A.; Morley, C.; Vandermeersch, T.; Hutchison, G. R. Open Babel: An open chemical toolbox. *Journal of Cheminformatics* **2011**, *3*.

- (22) Eberhardt, J.; Santos-Martins, D.; Tillack, A. F.; Forli, S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling* **2021**, *61*, 3891–3898.
- (23) Trott, O.; Olson, A. J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry* **2009**, *31*, 455–461.
- (24) Quiroga, R.; Villarreal, M. A. Vinardo: A Scoring Function Based on Autodock Vina Improves Scoring, Docking, and Virtual Screening. *PLOS ONE* **2016**, *11*, e0155183.